

Stochastic Gradient Descent - Main Result

$$\begin{aligned} \min: & f(x) \\ \text{st. } & x \in \mathcal{X} \end{aligned}$$

$$x \rightarrow \cancel{\nabla f(x)} \quad \cancel{g \in \partial f(x)}$$
$$\tilde{g}_x : \mathbb{E}[\tilde{g}_x] \in \partial f(x)$$

Thm: Suppose x^* exists, and $\mathbb{E}[\|\tilde{g}_x\|^2] \leq G^2 \quad \forall x$.
(compare to assumptions for sub-GD: $\|g_x\|^2 \leq G^2$)

SGD algorithm: $x_{t+1} = x_t - \eta \tilde{g}_x$

then: $\mathbb{E}\left[f\left(\frac{1}{T} \sum x_t\right)\right] - f(x^*) \leq \frac{R \cdot G}{\sqrt{T}}$

$$R^2 \geq \mathbb{E}[\|x_1 - x^*\|^2].$$

Convergence proof - 1/3

$$\mathbb{E} \left[\|x_{t+1} - x^*\|_2^2 \mid x_t \right] = \mathbb{E} \left[\|x_t - \eta \tilde{g}_t - x^*\|_2^2 \mid x_t \right]$$

$$= \mathbb{E} \left[\|x_t - x^*\|_2^2 - 2\eta \tilde{g}_t^\top (x_t - x^*) + \eta^2 \|\tilde{g}_t\|_2^2 \mid x_t \right]$$

$$\|x_t - x^*\|_2^2 - 2\eta \underbrace{\mathbb{E} [\tilde{g}_t \mid x_t]}_{g_t}^\top (x_t - x^*) + \eta^2 \mathbb{E} [\|\tilde{g}_t\|_2^2 \mid x_t]$$

$$\leq \|x_t - x^*\|_2^2 - 2\eta (f(x_t) - f(x^*)) + \eta^2 \mathbb{E} [\|\tilde{g}_t\|_2^2 \mid x_t]$$

Convergence proof - 2/3

Using the tower rule for conditional expectation

$$\begin{aligned} 0 &\leq \mathbb{E} \left[\|x_{t+1} - x^*\|_2^2 \right] \leq \mathbb{E} \left[\|x_t - x^*\|_2^2 \right] - 2\eta \left(\mathbb{E} f(x_t) - f(x^*) \right) \\ &\quad + \eta^2 \mathbb{E} \left[\|\tilde{g}_t\|_2^2 \right] \leq R^2 \\ &\leq \mathbb{E} \left[\|x_t - x^*\|_2^2 \right] - 2\eta \left(\mathbb{E} f(x_t) - f(x^*) \right) + \eta^2 G^2 \\ &\leq \mathbb{E} \left[\|x_1 - x^*\|_2^2 \right] - 2\eta \underbrace{\sum_{k=1}^t \left(\mathbb{E} f(x_k) - f(x^*) \right)}_{R^2} + \underbrace{t \eta^2 G^2}_{2\eta \cdot \sum_{k=1}^t \left(\mathbb{E} f(x_k) - f(x^*) \right)} \\ 2\eta \cdot \sum_{k=1}^t \left(\mathbb{E} f(x_k) - f(x^*) \right) &\leq R^2 + \eta^2 G^2 \end{aligned}$$

Convergence proof - 3/3

divide by 2η ; by $t \leftarrow T$

$$\underbrace{\mathbb{E} f\left(\frac{1}{T} \sum_{t=1}^T x_t\right) - f(x^*)}_{\text{}} \leq \frac{R^2}{2\eta T} + \frac{G^2 \eta}{2}$$

η that minimizes bound on RHS (take deriv, set to 0)

$$= \frac{R}{G} \cdot \frac{1}{\sqrt{T}}$$

using this :

$$\leq \frac{R G}{\sqrt{T}}$$

same convergence rate as
a function of T .

SGD for strongly convex functions

What happens when f is also strongly convex?

Thm: For all as above - $\mathbb{E}[\|\tilde{g}_t\|_2^2] \leq G^2$

If in addition f is α -strongly convex (may not be smooth), then

SGD with step decreasing step size: $\eta_t = \frac{2}{\alpha(t+1)}$

then:

$$f\left(\sum_{t=1}^T \frac{2t}{T(T+1)} \cdot x_t\right) - f(x^*) \leq \frac{2G^2}{\alpha} \cdot \frac{1}{T+1}$$

$\Rightarrow O\left(\frac{1}{T}\right)$ convergence rate.

Summary

- ① Same convergence rate as the deterministic case in the case where $\mathbb{E}[\|\tilde{g}_t\|^2]$ bdd:

$$O\left(\frac{1}{\sqrt{T}}\right) \quad (1/\varepsilon^2)$$

- ② same conv rate as the det. case when we also have strong convexity: $O(1/T)$. $(1/2)$

- ③ In deterministic setting: f is smooth & s.c.
 $\Rightarrow O(\exp\{-cT\}) \quad \log(1/\varepsilon)$

Can we get this?

Key to fast convergence rate:

self-tuning property

$$\nabla f(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow x^*$$

$\Rightarrow$ This allows us to take "big" step
size $\rightarrow \eta$ ind. of ε or T ($1/\rho$).

No self-tuning for SGD: $\mathbb{E} [\|\tilde{g}_n\|_0^2] \not\rightarrow 0$ as $n \rightarrow \infty$

Through picture:

